

Lecture 1

Results From QFT-1

Consider QFT of scalar field.

— We know that most of physical information is contained in

$$G(x_1, x_2, \dots, x_n) = \frac{\langle \Omega | T \left(\prod_{i=1}^N \hat{\phi}(x) \right) | \Omega \rangle}{\langle \Omega | \Omega \rangle}$$

where, $x = (\vec{x}, t)$

$|\Omega\rangle$ = interacting ground state.

$$T(\phi(t_1)\phi(t_2)) = \theta(t_1 - t_2)\phi(t_1)\phi(t_2) + \theta(t_2 - t_1)\phi(t_2)\phi(t_1)$$

(i.e. largest time on left)

The reason greens fun. is important is that it contains

— info. about masses of particles for a given theory (given by pole positions of 2 point functions in momentum space)

— It contains info about S -matrix elements. via LSZ formalism.

Goals For QFT-2 —

(1) One first goal in QFT-2 will be to develop new ways of calculating green's fun. ex. Path integral formalism & study properties of Green's fun.; we will also try to know why Path integral formulation? (1) Canonical Formulation breaks manifested lorentz invariance (we give special role to time in hamiltonian formulation i.e. Hamiltonian generates time translations not space translations); although final results are lorentz invariance, so one advantage is that we can maintain lorentz invariance in path integral formulation.

(2) Another reason to use path integral formulation is that, for cases of interaction term in lagrangian having time derivative than hamiltonian might become complicated ($H = p\dot{q} - L$)

Consider 1-d QM —

$$L = \frac{1}{2} \dot{q}^2 + \frac{\lambda}{2} \dot{q}^2 q$$

$$p = \frac{\partial L}{\partial \dot{q}} = \dot{q} + \lambda \dot{q} q$$

$$\Rightarrow \dot{q} = \frac{p}{1 + \lambda q}$$

$$\begin{aligned} H &= \dot{q} p - L \\ &= \frac{p^2}{(\lambda q + 1)} - \frac{1}{2} \frac{p^2}{(1 + \lambda q)^2} - \frac{\lambda}{2} q \frac{p^2}{(1 + \lambda q)^2} \\ &= \frac{p^2}{(1 + \lambda q)} \left(1 - \frac{1}{2(1 + \lambda q)} - \frac{q\lambda}{2(1 + \lambda q)} \right) \\ &= \frac{p^2}{2(1 + \lambda q)^2} (2 + 2\lambda q - 1 - \lambda q) \end{aligned}$$

$$H = \frac{1}{2} \left(\frac{p^2}{1 + \lambda q} \right) = \frac{p^2}{2} (1 + q\lambda)^{-1}$$

$$= \frac{p^2}{2} \left(1 - q\lambda + \frac{q^2 \lambda^2}{2} - \frac{\lambda^3 q^3}{3!} + \dots \right)$$

Note that lagrangian only contains one term of ‘ λ ’ whereas hamiltonian has multiple orders of λ .

(3) Path integral approach is well suited for non-abelian gauge theories. (non-abelian gauge theories describe strong, weak & electromagnetic interactions, i.e. everything except gravity).

We can also develop path integral approach for time dependent lagrangian but our lagrangian for universe is time independent upto great accuracy.

Path Integral in QM —

Lets start for simple example —

$$H = \frac{p^2}{2m} + V(q)$$

$$L = \frac{1}{2} m \dot{q}^2 - V(q)$$

we define Action $(S) = \int dt L(q(t), \dot{q}(t))$

Now consider quantum theory —

— We will use Heisenberg picture (time independent states & time dependent operator)

$$\Psi_H = \Psi_S(0), \quad O_H(0) = O_S$$

— set $\hbar = 1$

— Operators evolve via

$$\hat{q}(t) = e^{iHt} \hat{q}_H(0) e^{-iHt}$$

$$\hat{q}_H(t) = e^{iHt} \hat{q}_S e^{-iHt}$$

Now consider two states (eigen states of $\hat{q}$, labelled by e.v. q' & q''). $|q'\rangle$ & $|q''\rangle$ respectively.

Quantity we are interested in is $K(q', t'; q'', t'')$

$$K(q', t'; q'', t'') = \langle q'' | e^{-iH(t''-t')} | q' \rangle$$

$$= \langle q'' | \underline{e^{-iHt''}} \underline{e^{iHt'}} | q' \rangle$$

ex :

$$\hat{q}(t') \underline{e^{iHt'}} | q' \rangle = e^{iHt'} \hat{q} e^{-iHt'} e^{iHt'} | q' \rangle$$

$$= e^{iHt'} \hat{q} | q' \rangle$$

$$= q' e^{iHt'} | q' \rangle$$

So $e^{iHt'} | q' \rangle$ is eigen state of $\hat{q}(t')$ with e.v. q' .

Similarly,

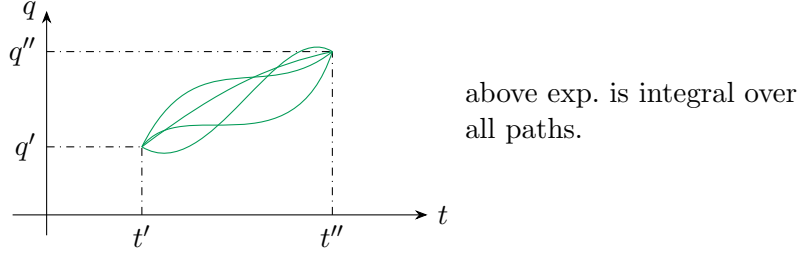
$$\langle q'' | e^{-iHt''} \hat{q}(t'') = q'' \langle q'' | e^{-iHt''}$$

So,

K is overlap between eigen states of $\hat{q}(t')$ & $\hat{q}(t'')$ with e.v. q' & q'' respectively.

Claim —

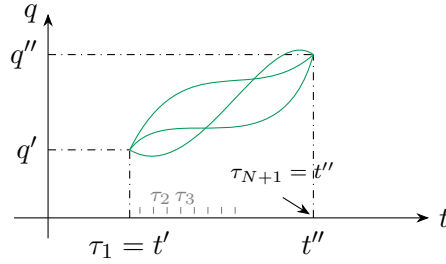
$$K(q', t'; q'', t'') = \int [\mathcal{D}q] e^{iS} \quad \text{for } t'' > t'$$



We think of path as discrete collection of points. Lets take N as some large integer.

We define $\Delta = \frac{t'' - t'}{N} \Rightarrow \text{small number}$

$$\begin{aligned}\tau_k &= t' + (k-1)\Delta \quad k = 1, 2, 3 \dots N+1. \\ \tau_1 &= t' \\ \tau_2 &= t' + \Delta \\ \tau_3 &= t' + 2\Delta \\ &\vdots \\ \tau_{N+1} &= t' + N\Delta = t''\end{aligned}$$



We define

$$\begin{aligned}q_i &= q(\tau_i) \quad \text{for a given path} \\ q_1 &= q(\tau_1) = q(t') = q' \\ q_2 &= q(\tau_2) \\ &\vdots \\ q_{N+1} &= q(\tau_{N+1}) = q(t'') = q''\end{aligned}$$

So our path is collection of points $(q_1, q_2 \dots q_{N+1})$ (or) $(q', \dots, q'')$

So action

$$\begin{aligned}(S) &= \int_{t'}^{t''} dt \left(\frac{1}{2} \dot{q}^2 - V(q) \right) \\ &= \Delta \sum_{k=1}^N \left(\frac{1}{2} \left(\frac{q_{k+1} - q_k}{\Delta} \right)^2 - V(q_k) \right)\end{aligned}$$

So

$$K(q', t'; q'', t'') = \lim_{N \rightarrow \infty} \mathcal{N} \int dq_2 dq_3 \dots dq_N e^{iS}$$

$\hookrightarrow$ does not depend on q' & q'' . (fixed for all case).

$$\begin{aligned} K(q', t'; q'', t'') &= \langle q'' | e^{-iH(t''-t')} | q' \rangle \\ &= \langle q'' | e^{-iHN\Delta} | q' \rangle \\ &= \langle q'' | \underbrace{e^{-iH\Delta}}_{\downarrow} e^{-iH\Delta} e^{-iH\Delta} \dots \underbrace{e^{-iH\Delta}}_{\uparrow} | q' \rangle \end{aligned}$$

$$\begin{aligned} &\int dq_N |q_N\rangle \langle q_N| \quad \text{insert completeness relation.} \\ &\int dq_2 |q_2\rangle \langle q_2| \\ &= \int dq_2 dq_3 \dots dq_N \prod_{k=1}^N \frac{\langle q_{k+1} | e^{-iH\Delta} | q_k \rangle}{\quad} \quad \left(\underline{q_{N+1} = q''} \text{ \& } q_1 = q' \right) \end{aligned}$$

Now our goal is to evaluate $\langle q_{k+1} | e^{-iH\Delta} | q_k \rangle$

$$\begin{aligned} &= \langle q_{k+1} | e^{-i\left(\frac{\hat{p}^2}{2m} + V(\hat{q})\right)\Delta} | q_k \rangle \quad \left| \begin{array}{l} e^{A+B} = e^A e^B e^{-\frac{1}{2}[B,A]} \dots \\ \text{for our case } [B, A] \sim \Delta^2 \end{array} \right. \\ &\simeq \langle q_{k+1} | e^{-i\frac{\hat{p}^2}{2m}\Delta} e^{-iV(\hat{q})\Delta} | q_k \rangle \end{aligned}$$

$\downarrow$ Note that this is approx. equal to.

$$\text{As } e^{A+B} = e^A e^B e^{\frac{1}{2}[B,A]} \dots$$

$$\text{For our case } [B, A] \sim \Delta^2 \Rightarrow e^{A+B} = e^A e^B \cdot e^{\Delta^2 K} = e^A e^B (1 + \Delta^2 K + \dots)$$

Since we have N -such products $\prod_{k=1}^N \langle q_{k+1} | e^{-iH\Delta} | q_k \rangle$ we get,

$$\begin{aligned} &\lim_{N \rightarrow \infty} (1 + \Delta^2 K)^N \quad \text{where } \Delta = C/N \\ &\lim_{N \rightarrow \infty} \left(1 + \frac{\text{const.}}{N^2} \right)^N = 1 \end{aligned}$$

$$\begin{aligned} &\simeq e^{-iV(q_k)\Delta} \langle q_{k+1} | e^{-i\frac{\hat{p}^2}{2m}\Delta} \cdot I | q_k \rangle \quad \hookrightarrow I = \int dp_k |p_k\rangle \langle p_k| \\ &\simeq e^{-iV(q_k)\Delta} \int dp \langle q_{k+1} | e^{-i\frac{\hat{p}^2}{2m}\Delta} | p_k \rangle \langle p_k | q_k \rangle \\ &\simeq e^{-iV(q_k)\Delta} \int dp e^{-i\frac{p_k^2}{2m}\Delta} \langle q_{k+1} | p_k \rangle \langle p_k | q_k \rangle \\ &\simeq e^{-iV(q_k)\Delta} \left(\frac{1}{2\pi} \right) \int dp e^{-i\frac{p_k^2}{2m}\Delta} \cdot e^{ip_k(q_{k+1}-q_k)} \end{aligned}$$

using

$$\begin{aligned}
\langle x|p\rangle &= \frac{1}{\sqrt{2\pi}} e^{+i p \cdot x} \\
&\simeq e^{-iV(q_k)\Delta} \left(\frac{1}{2\pi} \right) \int dp_k e^{-\frac{i\Delta}{2m} \left(p_k - \frac{2m p_k (q_{k+1}-q_k)}{\Delta} + \frac{m}{\Delta^2} (q_{k+1}-q_k)^2 - \frac{m}{\Delta^2} (q_{k+1}-q_k)^2 \right)} \\
&\simeq \frac{e^{-iV(q_k)\Delta}}{2\pi} \int dp_k e^{-\frac{i\Delta}{2m} \left(p_k - m \frac{(q_{k+1}-q_k)}{\Delta} \right)^2} e^{\frac{i}{2\Delta} (q_{k+1}-q_k)^2} \\
&\simeq \frac{e^{-iV(q_k)\Delta + \frac{i}{2\Delta} (q_{k+1}-q_k)^2}}{2\pi} \sqrt{\frac{\pi 2m}{i\Delta}}
\end{aligned}$$

So,

$$\begin{aligned}
K(q', t'; q'', t'') &= \int dq_2 dq_3 \dots dq_N \prod_{k=1}^N \left(\frac{1}{2\pi} \right) e^{-iV(q_k)\Delta + \frac{i}{2\Delta} (q_{k+1}-q_k)^2} \cdot \sqrt{\frac{2\pi m}{i\Delta}} \\
&= \int dq_2 dq_3 \dots dq_N \left(\frac{1}{\sqrt{2\pi}} \right)^N e^{+i\Delta \left(\sum_{k=1}^N \frac{(q_{k+1}-q_k)^2}{\Delta} - V(q_k) \right)} \left(\sqrt{\frac{m}{i\Delta}} \right)^N \\
\boxed{K(q', t'; q'', t'') &= \mathcal{N} \int [\mathcal{D}q] e^{+iS}} \quad \mathcal{N} = \left(\sqrt{\frac{m}{2\pi i\Delta}} \right)^N
\end{aligned}$$

Lecture 2

Generalisation :

Suppose we want to calculate matrix elements of kind —

$$\langle q'' | e^{-iHt''} T(\hat{q}(t_1) \hat{q}(t_2) \cdots \hat{q}(t_n)) e^{iHt'} | q' \rangle$$

instead of just $\langle q'', t'' | q', t' \rangle$ or $\langle q'' | e^{-iHt''} \cdot \mathbb{1} \cdot e^{iHt'} | q' \rangle$

where $t'' > t_n > t_{n-1} \cdots > t'$

eventually we will take limits $t'' \rightarrow \infty, t' \rightarrow -\infty$

Claim is

$$\langle q'' | e^{-iHt''} T(\hat{q}(t_1) \hat{q}(t_2) \cdots \hat{q}(t_n)) e^{iHt'} | q' \rangle = \int [\mathcal{D}q] e^{iS} q(t_1) q(t_2) \cdots q(t_n)$$

Please remind that $e^{iHt'} | q' \rangle$ is eigen state of $\hat{q}(t')$ as it satisfies

$$\begin{aligned} \hat{q}(t') e^{iHt'} | q' \rangle &= e^{iHt'} \hat{q} e^{-iHt'} e^{iHt'} | q' \rangle \\ &= q' e^{iHt'} | q' \rangle \end{aligned}$$

with eigen value q' .

Similarly $\langle q'' | e^{-iHt''}$ is eigen state of $\hat{q}(t'')$, with eigen value q'' .

Above claim implies that path integral automatically generates matrix elements of time ordered products of operators

Since LHS & RHS of our claim is independent of choice weather

$$\begin{aligned} t_1 > t_2 > t_3 \cdots > t_n \\ \text{or} \quad t_1 < t_2 < t_3 \cdots < t_{n-1} < t_n \end{aligned}$$

We choose $t_n > t_{n-1} > t_{n-2} \cdots > t_2 > t_1$

LHS :

$$\langle q'' | e^{-iHt''} \hat{q}(t_n) \hat{q}(t_{n-1}) \cdots \hat{q}(t_1) e^{iHt'} | q' \rangle$$

using

$$\left. \begin{aligned} \hat{q}(t_n) &= e^{iHt_n} \hat{q}(0) e^{-iHt_n} \\ &= e^{+iHt_n} \hat{q} e^{-iHt_n} \end{aligned} \right\} \text{Heisenberg picture.}$$

$\Rightarrow$ LHS becomes

$$\begin{aligned} &\langle q'' | e^{-iHt''} e^{iHt_n} \hat{q} e^{-iHt_n} e^{iHt_{n-1}} \hat{q} e^{-iHt_{n-1}} \cdots e^{iHt_1} \hat{q} e^{-iHt_1} e^{iHt'} | q' \rangle \\ &= \langle q'' | e^{-iH(t''-t_n)} \hat{q} e^{-iH(t_n-t_{n-1})} \cdots e^{-iH(t_2-t_1)} \hat{q} e^{-iH(t_1-t')} | q' \rangle \end{aligned}$$

Note that, time difference $t_n - t_{n-1}$ may not be same for all 'n'.

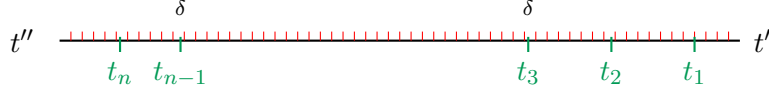
So let us descretise total time diff. $t'' - t' = N\Delta$

we define

$$\begin{aligned} t_1 &= t' + k_1 \Delta \\ t_2 &= t' + k_2 \Delta \quad \{k_r \text{ may not be same} \\ &\vdots \\ t_r &= t' + k_r \Delta \\ &\text{for some integer } k_r. \end{aligned}$$

Now one can argue that k_r may be some floating point number

The idea is that we divide $t'' - t'$ in large # of small time steps.



We then can shift our t_r such that it falls on one of red lines (upto an small order δ we can shift our times).

$$\text{Then} \quad t_r = t_1 + k_r \Delta \quad \text{where} \quad \underline{k_r \in \mathbb{I}}$$

$$r = 1, 2, 3 \dots n$$

$$k_r = 0, 1, 2, \dots N$$

Since we have choosen $t_n > t_{n-1} > t_{n-2} \dots > t_1$

$$\Rightarrow \quad t_r > t_{r-1} > t_{r-2}$$

$$\Rightarrow \quad k_r > k_{r-1} > k_{r-2}$$

$$\begin{aligned} \Rightarrow \quad t_n - t_{n-1} &= (t_1 + k_n \Delta) - (t_1 + k_{n-1} \Delta) \\ &= (k_n - k_{n-1}) \Delta \end{aligned}$$

$$\text{also} \quad t_1 = t' + k_1 \Delta \Rightarrow t_1 - t' = k_1 \Delta$$

LHS becomes :

$$\begin{aligned} &\langle q'' | e^{-i\hat{H}(t''-t_n)} \hat{q} e^{-i\hat{H}(t_n-t_{n-1})} \hat{q} \dots e^{-i\hat{H}(t_2-t_1)} \hat{q} e^{-i\hat{H}(t_1-t')} | q' \rangle \\ &= \langle q'' | e^{-iH(N-k_n)\Delta} \hat{q} e^{-iH(k_n-k_{n-1})\Delta} \dots e^{-iH(k_2-k_1)\Delta} \hat{q} e^{-iH k_1 \Delta} | q' \rangle \end{aligned}$$

$$\uparrow \quad \text{insert identity operator} \quad I = \int dq_n |q_n\rangle \langle q_n|$$

(Since $\exists$ 'n' such $\hat{q}$, we insert 'n' identity)

LHS becomes —

$$\begin{aligned} &= \langle q'' | e^{-i\hat{H}(N-k_n)\Delta} \hat{q} \left(\int dq_n |q_n\rangle \langle q_n| \right) e^{-i\hat{H}(k_n-k_{n-1})\Delta} \dots \\ &\quad \dots \hat{q} \int dq_2 |q_2\rangle \langle q_2| e^{-iH(k_2-k_1)\Delta} \hat{q} \int dq_1 |q_1\rangle \langle q_1| e^{-i\hat{H} k_1 \Delta} | q' \rangle \end{aligned}$$

$$\begin{aligned} \hat{q} |q_n\rangle &= q(t_n) |q_n\rangle \\ &= q_n |q_n\rangle \end{aligned}$$

LHS $\Rightarrow$

$$\begin{aligned} &\int dq_1 dq_2 \dots dq_n \langle q'' | e^{-i\hat{H}(N-k_n)\Delta} |q_n\rangle \langle q_n| e^{-i\hat{H}(k_n-k_{n-1})\Delta} |q_{n-1}\rangle \\ &\quad \dots \langle q_2 | e^{-i\hat{H}(k_2-k_1)\Delta} |q_1\rangle \langle q_1| e^{-i\hat{H} k_1 \Delta} | q' \rangle \\ &\quad \times q(t_1) q(t_2) \dots q(t_n) \end{aligned}$$

$$= \int dq_1 dq_2 \cdots dq_n \ q(t_1) q(t_2) \cdots q(t_n) \cdot \prod_{j=0}^n \langle q_{j+1} | e^{-iH(k_{j+1}-k_j)\Delta} | q_j \rangle$$

where,

$$q_{n+1} = q''$$

$$q_0 = q'$$

$$\langle q_{j+1} | e^{-iH(k_{j+1}-k_j)\Delta} | q_j \rangle = ?$$

or

$$\langle x | e^{-iH \alpha_j \Delta} | y \rangle = ? \quad \begin{cases} \alpha_j = k_{j+1} - k_j \\ q_j = y \\ q_{j+1} = x \end{cases}$$

$$\begin{aligned} & \langle x | e^{-i\left(\frac{\hat{p}^2}{2m} + \hat{V}(x)\right) \alpha_j \Delta} | y \rangle \\ & \sim \langle x | e^{-i\frac{V(x)}{2} \alpha_j \Delta} e^{-i\frac{\hat{p}^2}{2m} \alpha_j \Delta} e^{-iV(x) \frac{\alpha_j \Delta}{2}} | y \rangle \\ & \sim e^{-i\left(\frac{V(x)+V(y)}{2}\right) \alpha_j \Delta} \langle x | e^{-i\frac{\hat{p}^2}{2m} \alpha_j \Delta} I | y \rangle \quad \hookrightarrow I = \int dP |p\rangle \langle p| \\ & \sim e^{-i\left(\frac{V(x)+V(y)}{2}\right) \alpha_j \Delta} \int dP \langle x|p\rangle \langle p|y\rangle e^{-\frac{i p^2 \alpha_j \Delta}{2m}} \\ & \sim e^{-\frac{i}{2}(V(x)+V(y)) \alpha_j \Delta} \int dp \frac{1}{\sqrt{2\pi}} e^{ip \cdot x} \frac{1}{\sqrt{2\pi}} e^{-ip \cdot y} e^{-ip^2 \alpha_j \Delta / 2m} \\ & \sim \frac{1}{2\pi} e^{-\frac{i \alpha_j \Delta}{2}(V(x)+V(y))} \int dp e^{i\left(p \cdot (x-y) - \frac{p^2 \alpha_j \Delta}{2m}\right)} \\ & \sim \frac{1}{2\pi} e^{-\frac{i \alpha_j \Delta}{2}(V(x)+V(y))} \int dp e^{-\frac{i \alpha_j \Delta}{2m} \left(p^2 - \frac{2m}{\alpha_j \Delta} p \cdot (x-y) + \frac{m^2 (x-y)^2}{\alpha_j^2 \Delta^2} - \frac{m^2 (x-y)^2}{\alpha_j^2 \Delta^2}\right)} \\ & \sim \frac{1}{2\pi} e^{-i \frac{\alpha_j \Delta}{2}(V(x)+V(y))} \int dp e^{-i \frac{\alpha_j \Delta}{2m} \left(p - \frac{(x-y)m}{\alpha_j \Delta}\right)^2} e^{\frac{im(x-y)^2}{2\alpha_j \Delta}} \\ & \sim \frac{1}{2\pi} e^{-i \frac{\alpha_j \Delta}{2}(V(x)+V(y)) + \frac{im}{2\alpha_j \Delta} (x-y)^2} \int dp e^{-\frac{i \alpha_j \Delta}{2m} \left(p - \frac{(x-y)m}{\alpha_j \Delta}\right)^2} \\ & \sim \frac{1}{2\pi} e^{-i \frac{\alpha_j \Delta}{2}(V(x)+V(y)) + \frac{im}{2\alpha_j \Delta} (x-y)^2} \sqrt{\frac{\pi 2m}{\alpha_j \Delta}} \end{aligned}$$

$\Rightarrow$ LHS becomes,

$$\begin{aligned} & \int \mathcal{D}q \ q(t_1) q(t_2) \cdots q(t_n) \prod_{j=0}^n \frac{1}{2\pi} e^{-i \frac{\alpha_j \Delta}{2}(V(q_{j+1})+V(q_j)) + \frac{im}{2\alpha_j \Delta} (q_{j+1}-q_j)^2} \sqrt{\frac{\pi 2m}{\alpha_j \Delta}} \quad (\alpha = \alpha_j) \\ & = \mathcal{N} \int \mathcal{D}q \ q(t_1) q(t_2) \cdots q(t_n) e^{i\alpha \Delta \sum_{j=0}^n \frac{m}{2} \left(\frac{q_{j+1}-q_j}{\alpha \Delta}\right)^2 - \left(\frac{V(q_{j+1})+V(q_j)}{2}\right)} \end{aligned}$$

$$\boxed{\langle q'', t'' | T \left(\prod_i^n q(t_i) \right) | q', t' \rangle = \mathcal{N} \int \mathcal{D}q \ q(t_1) q(t_2) \cdots q(t_n) e^{iS}} \quad \begin{cases} \alpha \Delta = (k_{j+1} - k_j) \Delta \\ = t_{j+1} - t_j \end{cases}$$

Let us do the integral more accurately.

$$\int dp e^{-\frac{i\Delta p^2}{2m}}$$

to make a sense of it we need to add positive real part to 'i'.

$$\int dp e^{-(i+\epsilon)\frac{\Delta p^2}{2m}} \quad \left| \begin{array}{l} \text{For } i=0 \quad I = \text{Area} = \int dp e^{-\epsilon \Delta p^2 / 2m} \\ \text{Shall give } \text{+ve} \text{ answer} \end{array} \right.$$

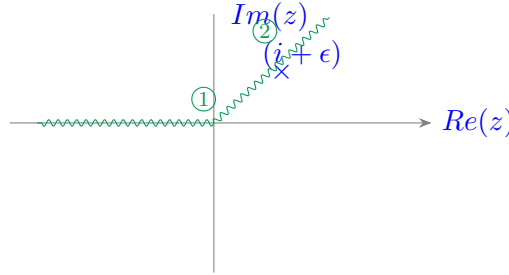

$$(\text{or}) \quad I = \int dp e^{-i(1-i\epsilon)\frac{\Delta p^2}{2m}}$$

Now apply gaussian formulæ

$$\begin{aligned} I &= \sqrt{\frac{\pi 2m}{(1-i\epsilon) i \Delta}} \\ &= \sqrt{\frac{2m \pi}{(i+\epsilon) \Delta}} \quad \leftarrow \text{Verify!} \end{aligned}$$

Time ordering & anti-time ordering is giving same answer in path integral reps, which it can't be — so we have to account for

Let us see how will we take $\sqrt{i+\epsilon}$ in this $\mathbb{C}$ plane. $\hookrightarrow$ has two solutions



If 'i' were absent, we should have positive answer. Branch cut has to go in some other direction.

Since we are defining analytic continuation from $\epsilon \rightarrow i+\epsilon$ i.e. there should not be branch cut in way of $\epsilon \rightarrow i+\epsilon$

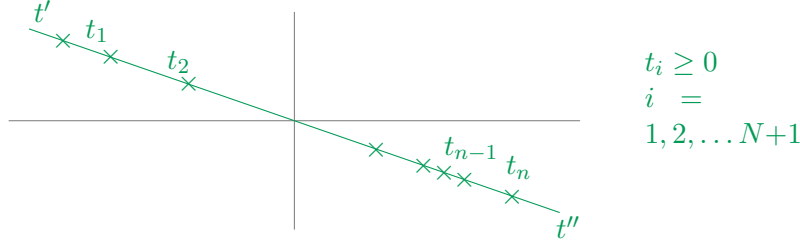
We can take branch cut like (1) or (2) does not matter.

$$\text{then} \quad \sqrt{1+\epsilon} = \lim_{\epsilon \rightarrow 0} (i+\epsilon)^{1/2} = e^{(i\pi/2)/2} = e^{i\pi/4}$$

This is same as,

$$\begin{aligned} \Delta &\longrightarrow \Delta(1-i\epsilon) \\ \frac{1}{\Delta} &\longrightarrow \frac{1}{\Delta}(1+i\epsilon) \\ \Delta &= \frac{t''-t'}{N} \end{aligned}$$

$$\begin{aligned} \text{for} \quad \Delta &\longrightarrow \Delta(1-i\epsilon) \\ t &\longrightarrow t(1-i\epsilon) \quad \hookrightarrow \text{complex time.} \end{aligned}$$



let us write $t = \tau (1 - i\epsilon)$ where $\tau \in \mathbb{R}, t \in \mathbb{C}$
 We have final path integral (for $\langle q(t'') | q(t') \rangle$)

$$\mathcal{N} \int dq_2 dq_3 \dots dq_n \exp \left(i\Delta \sum_{j=0}^N \left\{ \underbrace{\frac{m}{2} \left(\frac{q_{j+1} - q_j}{\Delta} \right)^2}_{\textcircled{1}} - \underbrace{V(q_j)}_{\textcircled{2}} \right\} \right)$$

here we have $\Delta \cdot \frac{1}{\Delta^2} \sim \frac{1}{\Delta}$

$$\frac{1}{\Delta} \rightarrow \frac{1}{\Delta} (1 + i\epsilon)$$

So extra term we get :

$$i \left(\frac{i\epsilon}{\Delta} \right) \sum_{j=1}^N \frac{m}{2} (q_{j+1} - q_j)^2 - i \sum_{j=0}^N (-i\epsilon) \Delta V(q_j)$$

For large q :

$$\int dq_2 dq_3 \dots dq_N \exp \left(-\frac{\epsilon}{\Delta} \sum_{j=0}^N \left(\frac{m}{2} (q_{j+1} - q_j)^2 \right) + i^2 \epsilon \sum_{j=0}^N V(q_j) \right)$$

for $q \rightarrow \infty$; $\exp(-\epsilon\infty) \rightarrow 0$

This extra term has effect that integral supresses for very large q . So This makes integral well defined; if it were purely oscilatory e^{iS} we remain in doubt if it is well defined or not.

Now we have a damped integral.

let us write $t = \tau (1 - i\epsilon)$; $\tau \in \mathbb{R}$

So,

$$\langle q'' | e^{-i\hat{H}\tau''(1-i\epsilon)} \underbrace{T \left(\hat{q}(\tau_1(1-i\epsilon)) \hat{q}(\tau_2(1-i\epsilon)) \dots \hat{q}(\tau_n(1-i\epsilon)) \right)}_{\text{}} e^{i\hat{H}\tau'(1-i\epsilon)} | q' \rangle$$

$$\downarrow I = \sum_s |s\rangle \langle s| \quad \hookrightarrow I = \sum_r |r\rangle \langle r| \quad \{|r\rangle\} \text{ are energy eigen states}$$

let us insert complete set of energy eigen states —

$$I = \sum_s |s\rangle \langle s|$$

Now term

$$1) \quad \langle q'' | \sum_s |s\rangle \langle s| e^{-i\hat{H}\tau''(1-i\epsilon)} = \sum_s \langle q'' | s \rangle e^{-i E_s \tau''(1-i\epsilon)} \langle s| \\ = \sum_s \langle q'' | s \rangle e^{-i E_s \tau'' - \epsilon E_s \tau''} \langle s|$$

$$2) \quad e^{i\hat{H}\tau'(1-i\epsilon)} \sum_r |r\rangle \langle r| q' \rangle = \sum_r \langle r| q' \rangle e^{i E_r \tau'(1-i\epsilon)} |r\rangle \\ = \sum_r \langle r| q' \rangle e^{+i E_r \tau' + \epsilon E_r \tau'} |r\rangle$$

Now; take limits : $\tau'' \rightarrow \infty$, $\tau' \rightarrow -\infty$

under these limits term ① & ② only survives for $E_r = E_s = E_\Omega$ i.e ground state.

$$\langle q'' | \Omega \rangle \langle \Omega | q' \rangle e^{-i E_\Omega \tau'' - \epsilon E_\Omega \tau''} e^{i E_\Omega \tau' + \epsilon E_\Omega \tau'} \quad \nwarrow \text{ground state (vacuum)}$$

Then

$$\langle \Omega | T(\hat{q}(\tau_1(1-i\epsilon)) \hat{q}(\tau_2(1-i\epsilon)) \cdots \hat{q}(\tau_n(1-i\epsilon))) | \Omega \rangle \quad \downarrow \langle q(\tau'' \rightarrow \infty) | \\ \hookrightarrow | \hat{q}(\tau' \rightarrow -\infty) \rangle$$

Normalising :

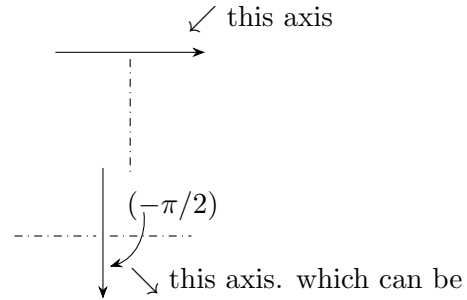
$$\frac{\langle \Omega | T(\prod_{i=1}^n \hat{q}(\tau_i(1-i\epsilon))) | \Omega \rangle}{\langle \Omega | \Omega \rangle} = \frac{\int [\mathcal{D}q] e^{iS} q(\tau_1(1-i\epsilon)) q(\tau_2(1-i\epsilon)) \cdots q(\tau_n(1-i\epsilon))}{\int [\mathcal{D}q] e^{iS}}$$

{in end we take $\epsilon \rightarrow 0$ limit.}

Since we are rotating time axis $t \rightarrow \tau(1-i\epsilon)$ we can rotate it all the way to imaginary axis.

i.e. instead of working on

we work on



achieved by $t \rightarrow e^{-i\pi/2} \tau$

Problem is argument of $q(\tau e^{-i\pi/2})$ are imaginary; we will do calculation using these arguments and will rotate the final result back to where arguments are real.

i.e. we will have to rotate $e^{+i\pi/2}$ after our final result. $\hookrightarrow$ anticlockwise

If we find result for anti-time ordering; then this will be different. $\Delta \rightarrow i\Delta = \tau$ i.e. $t_2 - i\tau_2$

Singularities appear when we do analytic continuation back to real axis. There are also studies on kinds of singularity which appear on analytic continuation.

Anti-time ordering gives us euclidean metric path integral by rotating $t \rightarrow e^{i\pi/2} \tau$; but we won't use it; since we are dealing with perturbative QFT. Euclidean path integral provides results for non-perturbative phenomena.

The funda is same; after we found results using euclidean path integral we will rotate back to real axis. $(t \rightarrow e^{-i\pi/2} \tau)$ where $(\tau' = e^{i\pi/2} \tau)$.

If vacuum were degenerate; this would have been more complicated. Fortunately in most of QFT vacuum is non-degenerate.

In QM if there is degenerate vacuum; we have to be more careful.

Lecture 3

Scalar field theory :

$$S = \int dt d^3x \left(-\frac{1}{2} \eta^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - V(\phi) \right)$$

here, $\eta^{\mu\nu} = \text{diag}(-1, 1, 1, 1)$

Think of space as discrete collection of points.

$$\vec{x} \longrightarrow a \vec{n} \quad \vec{n} = (n_1, n_2, n_3)$$

↓ lattice size

$$\text{Space} \longrightarrow \begin{array}{ccccccc} & \cdot & \cdot & \cdot & \cdot & \cdot & \\ & \cdot & \cdot & \cdot & \cdot & \cdot & \\ & \cdot & \cdot & \cdot & \cdot & \cdot & \end{array}$$

using discretised space breaks lorentz invariance. we will use it in limit $a \rightarrow 0$.

$$\begin{aligned} \text{Action } (S) &= \int dt d^3x \left(\frac{\dot{\phi}^2}{2} - \frac{(\nabla \phi)^2}{2} - V(\phi) \right) & \phi(x, t) &\rightarrow \phi_n(t) \\ &= \int dt a^3 \sum_{n=(n_1, n_2, n_3)} \left[\left(\frac{1}{2} \partial_t \phi_n(t) \right)^2 \right. \\ &\quad - \frac{1}{2} \left(\frac{\phi_{n_1+1, n_2, n_3}(t) - \phi_{n_1 n_2 n_3}(t)}{a} \right)^2 \\ &\quad - \frac{1}{2} \left(\frac{\phi_{n_1, n_2+1, n_3}(t) - \phi_{n_1 n_2 n_3}(t)}{a} \right)^2 \\ &\quad - \frac{1}{2} \left(\frac{\phi_{n_1, n_2, n_3+1}(t) - \phi_{n_1 n_2 n_3}(t)}{a} \right)^2 \\ &\quad \left. - V(\phi) \right] \end{aligned}$$

Exercise : check the path integral formulation extends to multiple variables.

$$H = \sum_{i=1}^M \frac{p_i^2}{2m_i} + V(q_1, q_2 \dots q_M)$$

Discretize time with interval Δ .

$$\text{i.e.} \quad \int dt \longrightarrow \Delta \sum_{n_0} \quad n = (n_0, n_1, n_2, n_3)$$

here, $n_0, n_1, n_2, n_3 \in \text{Integers}$

$$\begin{aligned} \partial_t \phi &\longrightarrow \frac{\phi_{n_0+1, n_1, n_2, n_3} - \phi_{n_0, n_1, n_2, n_3}}{\Delta} \\ \phi_{n_1 n_2 n_3}(t) &\longrightarrow \phi_n = \phi_{n_0, n_1, n_2, n_3} \end{aligned}$$

The $\Delta \rightarrow \Delta(1 - i\epsilon)$ prescription is same as working in Euclidean prescription.

if Δ is real time

$$\Delta \longrightarrow \tau = i \Delta \quad \left\{ \begin{array}{c} \nearrow^\tau \\ \Delta \end{array} \right\} \quad \text{here } \tau \text{ is imaginary time.}$$

If we were in Euclidean formulation $(-i\tau = \Delta \Rightarrow \tau = i\Delta)$

$$\begin{aligned} \partial_t \phi &\longrightarrow \partial_\tau \phi \\ \frac{\phi_{n_0+1, n_1 n_2 n_3} - \phi_{n_0 n_1 n_2 n_3}}{\Delta} &\longrightarrow \frac{\phi_{n_0+1, n_1 n_2 n_3} - \phi_{n_0 n_1 n_2 n_3}}{-i \tau} \end{aligned}$$

So,

$$(\partial_\tau \phi)^2 = -(\partial_t \phi)^2$$

For $\Delta = a$; space & time are symmetrical

Euclidean formulation makes it more manifest that there is some symmetry in $x \leftrightarrow t$.

Once we take out continuum limit, we get a continuous rotation group, and then we do analytic continuation, to recover lorentz symmetry.

$$\begin{aligned} S[\phi] = \text{Functional of } \phi(x) &= \int d^4x \left(\frac{1}{2} (\partial_t \phi)^2 - \frac{1}{2} (\nabla \phi)^2 - V(\phi(x, t)) \right) \\ x &= (\bar{x}, t) \end{aligned}$$

So,

$S[\phi]$: Functional of ϕ . in continuum formalism but it is a function of $\{\phi_{i_1 i_2 i_3 i_4}\}$ in discrete form.

$$\begin{aligned} \{\phi_i\} &= \{\phi_{0000}, \phi_{0001}, \phi_{0011}, \dots, \phi_{(N-1)(N-1)(N-1)(N-1)}\} \\ &\hookrightarrow \text{variables.} \end{aligned}$$

So we can define usual partial derivative — $\frac{\partial S}{\partial \phi_{i_1 i_2 i_3 i_4}}$

(Defⁿ) Functional derivative $\xrightarrow{\text{discretisation}}$

$$\frac{1}{a^3 \Delta} \frac{\partial S}{\partial \phi_{i_0 i_1 i_2 i_3}} \equiv \frac{\delta S}{\delta \phi(x)}$$

Claim :

For fixed y , $\phi(y)$ is a functional of $\phi(x)$.

Since $\phi(y)$ gives a number for a given $\phi(x)$ at point 'y'.

So, we can calculate,

$$\text{functional derivative} \quad \frac{\delta \phi(y)}{\delta \phi(x)} = \delta^4(x - y)$$

Proof :

Suppose : $y = (\Delta n_0, a n_1, a n_2, a n_3)$

$$\begin{aligned} \phi(y) &= \phi_{n_0 n_1 n_2 n_3} \\ \frac{\delta \phi(y)}{\delta \phi(x)} &= \frac{1}{a^3 \Delta} \frac{\partial \phi_{n_0 n_1 n_2 n_3}}{\partial \phi_{m_0 m_1 m_2 m_3}} \\ &= \frac{1}{a^3 \Delta} \delta_{m_0 n_0} \delta_{m_1 n_1} \delta_{m_2 n_2} \delta_{m_3 n_3} \end{aligned}$$

Lets prove RHS to be equal to $\delta^4(x - y)$.

$$\int d^4x \delta^4(x - y) F(x) = F(y) \quad \text{for any } F.$$

↓

$$\begin{aligned} a^3 \Delta \sum_{m_0 m_1 m_2 m_3} \frac{1}{a^3 \Delta} \delta_{m_0 n_0} \delta_{m_1 n_1} \delta_{m_2 n_2} \delta_{m_3 n_3} F_{m_0, m_1 m_2 m_3} \\ = F_{n_0 n_1 n_2 n_3} = F(y) \end{aligned}$$

This has to hold for any F .

In scalar Field Theory :

$$\begin{aligned} Z[J] &= \int \mathcal{D}\phi \ e^{iS(\phi) + i \int d^4x J(x) \phi(x)} \\ &\quad (\text{Functional of } J \text{ \& Not } \phi) ! (?) \\ \left(-i \frac{\partial}{\partial J(x_1)}\right) \left(-i \frac{\partial}{\partial J(x_2)}\right) \cdots \left(-i \frac{\partial}{\partial J(x_n)}\right) Z[J] &= Z[J] \phi(x_1) \phi(x_2) \cdots \phi(x_n) \\ \left(\text{using } \frac{d}{dx} e^{a(x)} = e^{a(x)} \frac{da}{dx}\right) & \end{aligned}$$

So,

$$\begin{aligned} -i \frac{\partial}{\partial J(x_n)} \int \mathcal{D}\phi \ e^{iS(\phi) + i \int J(x) \phi(x) d^4x} &= (-i) (i) Z \cdot \frac{\partial}{\partial J(x_n)} \int J(x) \phi(x) d^4x \\ &= Z[J] \cdot \int \frac{\partial}{\partial J(x_n)} J(x) \phi(x) d^4x \\ &= Z[J] \int \delta^4(x - x_n) \phi(x) d^4x \\ &= Z[J] \phi(x_n) \end{aligned}$$

$$\begin{aligned} -i \frac{\partial}{\partial J(x_{n-1})} Z[J] \phi(x_n) &= Z[J] \phi(x_{n-1}) \phi(x_n) \\ -i \frac{\partial}{\partial J(x_2)} Z[J] \phi(x_{n-1}) \phi(x_n) &= Z[J] \phi(x_2) \phi(x_{n-1}) \phi(x_n) \end{aligned}$$

So,

$$\boxed{\left(\prod_{i=1}^n \left(-i \frac{\partial}{\partial J(x_i)}\right)\right) Z[J] = Z[J] \prod_{i=1}^n \phi(x_i)} \quad \longleftarrow \text{classical fields/variables}$$

Similarly,

$$G_n(x_1, x_2 \dots x_n) = \frac{\langle \Omega | T \left(\overbrace{\hat{\phi}(x_1) \hat{\phi}(x_2) \dots \hat{\phi}(x_n)}^{\text{operators.}} \right) | \Omega \rangle}{\langle \Omega | \Omega \rangle}$$

using;

$$\langle q'' | e^{-iHt''} \underline{T(\hat{q}(t_1) \hat{q}(t_2) \cdots \hat{q}(t_n))} e^{iHt'} | q' \rangle = \int [\mathcal{D}q] e^{iS} q(t_1) q(t_2) \cdots q(t_n)$$

$$= \frac{1}{Z[0]} \left(-i \frac{\partial}{\partial J(x_1)} \right) \left(-i \frac{\partial}{\partial J(x_2)} \right) \cdots \left(-i \frac{\partial}{\partial x_n} \right) Z[J] \Big|_{J=0}.$$

$\uparrow$ (classical variables)

Lecture 4

Calculate $Z[J]$ for free theory :

$$Z[J] = \int \mathcal{D}\phi \ e^{iS(\phi) + i \int J(x) \phi(x) d^4x} \quad \text{---(1)}$$

$$\eta_{\mu\nu} = (-1, 1, 1, 1)$$

$$S[\phi]_{\text{free}} = \int d^4x \left(-\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 \right)$$

Lets find $(iS[\phi] + i \int J(x) \phi(x) d^4x)$ first then put it in eq ①.

Claim is: $iS[\phi] + i \int J(x) \phi(x) d^4x$ is equal to

$$\frac{i}{2} \int d^4x (-\eta^{\mu\nu} \partial_\mu \chi \partial_\nu \chi - m^2 \chi^2) - \frac{1}{2} \int d^4x d^4x' \Delta(x-x') J(x) J(x')$$

Where $\Delta(x-x')$ satisfies

$$(\partial_x^2 - m^2) \Delta(x-x') = i \delta^4(x-x')$$

&

$$\chi(x) = \phi(x) - i \int d^4x' \Delta(x, x') J(x')$$

RHS :

$$\begin{aligned} & \frac{i}{2} \int d^4x (-\partial_\mu \chi \partial^\mu \chi - m^2 \chi^2) \quad \left\{ \begin{array}{l} \text{using } \int \partial_x (\chi \partial^\mu \chi) = \underbrace{(\chi \partial^\mu \chi)}_{\text{surface term.}} \\ = 0 = \int \partial_\mu \chi \partial^\mu \chi + \int \chi \partial_x^2 \chi \\ \chi(\text{Boundary}) \rightarrow 0 \end{array} \right. \\ &= \frac{i}{2} \int d^4x (\chi \partial^2 \chi - m^2 \chi^2) \\ &= \frac{i}{2} \int d^4x \chi (\partial_x^2 - m^2) \chi \\ &= \frac{i}{2} \int d^4x \chi(x) (\partial_x^2 - m^2) \left(\phi(x) - i \int d^4x' \Delta(x-x') J(x') \right) \\ &= \frac{i}{2} \int d^4x \chi(x) (\partial_x^2 - m^2) \phi(x) + \frac{1}{2} \int d^4x d^4x' \chi(x) \underbrace{(\partial_x^2 - m^2) \Delta(x-x')} J(x') \\ &= \frac{1}{2} \int d^4x d^4x' \chi(x) \delta^4(x-x') J(x') + \frac{i}{2} \int d^4x \chi(x) (\partial_x^2 - m^2) \phi(x) \\ &= \frac{i}{2} \int d^4x \chi(x) (\partial_x^2 - m^2) \phi(x) + \frac{1}{2} \int d^4x \chi(x) J(x) \end{aligned}$$

①
②

using integral by parts $\leftrightarrow$ we get

$$\begin{aligned}
&= \frac{i}{2} \int d^4x \phi (\partial_x^2 - m^2) \chi(x) + \frac{1}{2} \int d^4x \left(\phi(x) - i \int d^4x' \Delta(x-x') J(x') \right) J(x) \\
&= \frac{i}{2} \int d^4x \phi(x) (\partial_x^2 - m^2) \left\{ \phi - i \int d^4x' \Delta(x-x') J(x') \right\} \\
&\quad + \frac{1}{2} \int d^4x \phi(x) + \frac{1}{2} \int d^4x d^4x' \Delta(x-x') J(x') J(x) \\
&= \frac{i}{2} \int d^4x \phi (\partial_x^2 - m^2) \phi + \frac{1}{2} \int d^4x d^4x' \phi(x) \underbrace{(\partial_x^2 - m^2) \Delta(x-x')}_{\nearrow i \delta^4(x-x')} J(x') \\
&\quad + \frac{i}{2} \int d^4x \phi(x) + \frac{1}{2} \int d^4x d^4x' \Delta(x-x') J(x') J(x) \\
&= \frac{i}{2} \int d^4x \phi (\partial_x^2 - m^2) \phi + \frac{i}{2} \int d^4x \phi(x) J(x) + \frac{i}{2} \int d^4x \phi(x) J(x) \\
&\quad + \frac{1}{2} \int d^4x d^4x' \Delta(x-x') J(x') J(x)
\end{aligned}$$

Now,

$$\begin{aligned}
&\frac{1}{2} \int d^4x (-\partial_\mu \phi \partial^\mu \phi - m^2 \phi^2) - \frac{1}{2} \int d^4x d^4x' \Delta(x-x') J(x) J(x') \\
&= \frac{i}{2} \int d^4x \phi (\partial_x^2 - m^2) \phi + i \int d^4x \phi(x) J(x) \\
&\quad + \frac{1}{2} \int d^4x d^4x' \Delta(x-x') J(x') J(x) - \frac{1}{2} \int d^4x d^4x' \Delta(x-x') J(x') J(x) \\
&= \frac{i}{2} \int d^4x \phi (\partial_x^2 - m^2) \phi + i \int d^4x \phi(x) J(x) \\
&= i S[\phi] + i \int d^4x \phi(x) J(x) = \underline{\text{LHS}}. \quad \text{proved!} \\
&\Rightarrow i \int d^4x \phi(x) J(x) = -\frac{1}{2} \int d^4x d^4x' \Delta(x-x') J(x) J(x')
\end{aligned}$$

Idea is now to replace $\mathcal{D}\phi \rightarrow \mathcal{D}\chi$
as ϕ is just a constant shift to χ .

$$\begin{aligned}
\chi &= \phi - i \int d^4x' \Delta(x-x') J(x') \\
d\chi &= d\phi \\
\Rightarrow \mathcal{D}\phi &= \mathcal{D}\chi
\end{aligned}$$

So,

$$\begin{aligned}
Z[J] &= \int \mathcal{D}\chi e^{iS(\chi) + i \int J(x) \chi(x) d^4x} \\
&= \int \mathcal{D}\chi e^{\frac{i}{2} \int d^4x \chi (\partial_x^2 - m^2) \chi} \cdot e^{-\frac{1}{2} \int d^4x d^4x' \Delta(x-x') J(x) J(x')}
\end{aligned}$$

$$Z_f[J] = Z_f[0] e^{-\frac{1}{2} \int d^4x d^4x' \Delta(x-x') J(x) J(x')}$$

$$\Rightarrow \boxed{\frac{Z_f[J]}{Z_f[0]} = \exp\left(-\frac{1}{2} \int d^4x d^4x' \Delta(x-x') J(x) J(x')\right)}$$

With this we can find free greens function —

$$G_f(x_1, x_2 \dots x_n) = \left(\frac{-i \partial}{\partial J(x_1)} \right) \left(\frac{-i \partial}{\partial J(x_2)} \right) \dots \left(\frac{-i \partial}{\partial J(x_n)} \right) \frac{Z_f[J]}{Z_f[0]} \Big|_{J=0}.$$

example,

$$\begin{aligned} G_f(x_1 x_2) &= \left(-\frac{i \partial}{\partial J(x_1)} \right) \left(\frac{-i \partial}{\partial J(x_2)} \right) \frac{Z_f[J]}{Z_f[0]} \Big|_{J=0} \\ &= \left(\frac{-i \partial}{\partial J(x_1)} \right) \left(\frac{-i \partial}{\partial J(x_2)} \right) \cdot e^{-\frac{1}{2} \int d^4x d^4x' \Delta(x-x') J(x) J(x')} \\ &= \left(-\frac{i \partial}{\partial J(x_1)} \right) e^{-\frac{1}{2} \int d^4x d^4x' J(x) \Delta(x-x') J(x')} \\ &\quad \cdot \left(\frac{i}{2} \int d^4x d^4x' J(x) \Delta(x-x') \frac{\partial J(x')}{\partial J(x_2)} + \frac{i}{2} \int d^4x d^4x' \frac{\partial J(x)}{\partial J(x_2)} \Delta(x-x') J(x') \right) \\ &= \left(-\frac{i \partial}{\partial J(x_1)} \right) e^{-\frac{1}{2} \int d^4x d^4x' \Delta(x-x') J(x) J(x')} \cdot \left(\frac{i}{2} \int d^4x d^4x' J(x) \Delta(x-x') \cdot \delta^4(x' - x_2) \right. \\ &\quad \left. + \frac{i}{2} \int d^4x d^4x' J(x') \Delta(x-x') \delta^4(x - x_2) \right) \\ &= \left(-\frac{i}{\partial J(x_1)} \right) e^{-\frac{1}{2} \int d^4x d^4x' J(x) \Delta(x-x') J(x')} \left\{ \frac{i}{2} \int d^4x J(x) \Delta(x - x_2) + \frac{i}{2} \int d^4x' J(x') \Delta(x_2 - x') \right\} \\ &= \left(\frac{i}{2} \int d^4x d^4x' \frac{\delta J(x)}{\delta J(x_1)} \Delta(x-x') J(x') \right. \\ &\quad \left. + \frac{i}{2} \int d^4x d^4x' J(x) \Delta(x-x') \frac{\delta J(x')}{\delta J(x_1)} \right) \times \left(\frac{1}{2} \int d^4x \frac{J(x)}{\Delta(x-x')} + \frac{1}{2} \int d^4x' J(x') \Delta(x_2 - x') \right) \\ &\quad + e^{-\frac{1}{2} \int d^4x d^4x' J(x) \Delta(x-x') J(x')} \left(\frac{1}{2} \int d^4x \delta^4(x - x_1) \Delta(x - x_2) \right. \\ &\quad \left. + \frac{1}{2} \int d^4x' \delta^4(x' - x_1) \Delta(x_2 - x') \right) \end{aligned}$$

at $J = 0$

$$\begin{aligned} &= \frac{1}{2} \Delta(x_1 - x_2) + \frac{1}{2} \Delta(x_2 - x_1) \\ &= \Delta(x_1 - x_2) = \Delta(x_1, x_2) \end{aligned}$$

Similarly,

$$\begin{aligned} G_{\text{free}}(x_1 x_2 x_3 x_4) &= \Delta(x_1 x_2) \Delta(x_3 x_4) + \Delta(x_1 x_3) \Delta(x_2 x_4) \\ &\quad + \Delta(x_1 x_4) \Delta(x_2 x_3) \end{aligned}$$

Similarly,

$$G_{\text{free}}(x_1 x_2 \dots x_{2n}) = \Delta(x_1 x_2) \Delta(x_2 x_3) \dots \Delta(x_{2n-1}, x_{2n}) + \text{All permutations}$$

$$\text{Total \# of permutations} = \frac{2n!}{2^n n!}$$

as

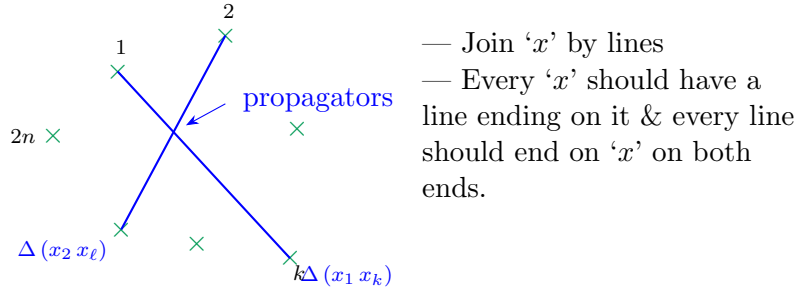
$$G_f(x_1 x_2 \dots x_{2n+1}) = 0 \quad (\text{using wicks contraction } \langle 0|T(\text{odd \# of fields})|0\rangle = 0)$$

(or) # of permutations shall be :

$$(2n-1)(2n-3) \cdot 5 \cdot 3 \cdot 1 = \frac{2n!}{2^n n!} \quad \leftarrow \text{How ?}$$

Feynman rules :

(1) For correlation function of $2n$ - ϕ 's



To get $\Delta(x_1 x_2 \dots x_{2n})$ we find :

$$\Delta(x_1 x_2 \dots x_{2n}) = \underbrace{\Delta(x_1, x_k) \Delta(x_2, x_\ell) \dots}_{\substack{\text{'n' such terms} \\ \text{(n- propagators)}}} + \underbrace{\sum}_{n} \text{All permutations}$$

Suppose we take action —

$$S = - \int d^4x \left(\frac{1}{2} \eta^{\mu\nu} \partial_\mu \phi \partial_\nu \phi + \frac{m^2}{2} \phi^2 + \frac{\lambda}{4!} \phi^4 \right)$$

λ : small parameter

Write

$$\exp \left(\int \left(i \frac{\lambda}{4!} \phi^4 \right) d^4x \right) \text{ as : } \sum_{m=0}^{\infty} \frac{1}{m!} \left(\frac{-i\lambda}{4!} \right)^m \left\{ \int d^4x \phi^4(x) \right\}^m$$

$$e^{iS} = e^{iS_{\text{free}} + iS_{\text{int}}} = e^{iS_{\text{free}}} \cdot e^{iS_{\text{int}}} = e^{iS_{\text{free}}} \sum_m \frac{1}{m!} \left(\frac{-i\lambda}{4!} \right)^m \{iS_{\text{int}}\}^m$$

We are interested to find $G(x_1 x_2 \dots x_{2n})$ in interacting theory.

$$\text{unnormalised } G(x_1 x_2 \dots x_n) = \langle 0|T \{ \phi_0(x_1) \phi_0(x_2) \dots \phi_0(x_n) e^{iS_{\text{int}}} \} |0\rangle$$

$$= \langle 0|T \left\{ \phi_0(x_1) \phi_0(x_2) \dots \phi_0(x_n) \sum_{m=0}^{\infty} \frac{1}{m!} \left(\frac{-i\lambda}{4!} \right)^m \left(\int d^4y \phi^4(y) \right)^m \dots \right\} |0\rangle$$

$$\text{Normalised } G(x_1 x_2 \dots x_n) = \frac{G(x_1 x_2 \dots x_n)}{G(x_1 x_2 \dots x_n|_{n=0})} = \frac{G(x_1 x_2 \dots x_n)}{G(0)}$$

$$G(0) = \langle \Omega | \Omega \rangle = \langle 0 | T \{ \exp(i S_{\text{int}}) \} | 0 \rangle$$

Feynman rules of interacting theory :

Besides the x 's for external states we have 4 extra x 's for each factor of λ .

i.e. For each order in ' λ ' introduce a vertex  & calculate using 4 extra vertex in free case with a vertex factor of $\left(\frac{-i\lambda}{4!}\right)$.

$$\text{and} \quad \int d^4 x \quad \forall x.$$

For m such vertex  we have $\frac{1}{m!}$ from expansion of $e^{i S_{\text{int}}}$.

or $\left(\text{for } O(\lambda^m) \text{ we have } \frac{1}{m!} \text{ factor.}\right)$

Suppose we want to calculate $G(x_1 x_2)$ upto $O(\lambda)$.

$$\begin{array}{c} 1 \times \\ \\ 2 \times \end{array} \quad \begin{array}{c} \diagup \diagdown \\ \times \end{array} \quad \downarrow \text{ one vertex with factor } \left(-\frac{i\lambda}{4!}\right)$$

Possibilities are:

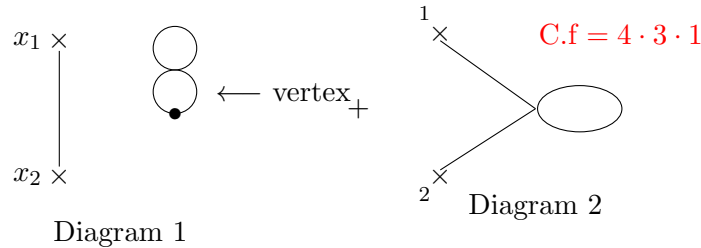


Diagram (1) : ϕ^4 theory

$$G_1 = \left(\frac{-i\lambda}{4!}\right) \int d^4 x \Delta(x_1 x_2) \Delta(x, x) \Delta(x, x) \cdot (3) \quad \checkmark \text{ Combinatorial factor}$$

Diagram (2) :

$$G_2 = \frac{-i\lambda}{4!} \int d^4 x \Delta(x_1, x) \Delta(x_2, x) \Delta(x, x) \cdot (4 \cdot 3 \cdot 1)$$

$$G(x_1 x_2) = G_1 + G_2$$

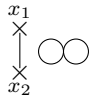
ex :

$$G(x_1 x_2 x_3 x_4) = \begin{array}{c} x_1 \quad x_2 \\ \diagdown \quad \diagup \\ x \\ \diagup \quad \diagdown \\ x_3 \quad x_4 \end{array} = \frac{-i\lambda}{(4!)} \int d^4 x \Delta(x_1, x) \Delta(x_2, x) \Delta(x_3, x) \Delta(x_4, x) \cdot (4 \cdot 3 \cdot 2 \cdot 1)$$

$\checkmark$ $4!$ is designed to cancel out C.f. from tree level diagrams! $\downarrow$ Combinatorial factor

denominator / Normalisation

$$\begin{aligned}\langle\Omega|\Omega\rangle &= \langle 0|T\{e^{iS_{\text{int}}}\}|0\rangle \\ &= 1 + \int \left(\frac{-i\lambda}{4!}\right) d^4x \langle 0|\phi^4(x)|0\rangle + \left(\frac{-i\lambda}{4!}\right)^2 \frac{1}{2!} \int d^4x d^4y \langle 0|\phi^4(x)\phi^4(y)|0\rangle + \dots \\ &= 1 + \text{bubble} + \text{two bubbles} + \text{self-energy} + \dots \text{Bubble diagrams}\end{aligned}$$

diagrams like  should not be included as such disconnected diagrams factor as connected $\times$ vacuum bubbles where bubbles's contribution cancels from denominator $\langle\Omega|\Omega\rangle$.

$$\text{-----} \times \text{-----}$$

Lecture 5 — Factorisation of bubble diagrams

Factorisation of bubble diagrams —

Suppose we have situation like this :



(m – interaction vertices)

Feynman diagram with no bubble (as we have external legs)

(n -vertices [all are internal])

vacuum Bubble (as no external line/leg)

Interpretation :

$$\frac{1}{(m+n)!} \left(\frac{-i\lambda}{4!} \right)^{m+n} \quad \checkmark$$

(1) Above two diagrams may appear as single feynman diagram

(2) Above diagram is product of two separete diagrams. (connected $\times$ bubble).

$$= \frac{1}{m!} \left(\frac{-i\lambda}{4!} \right)^m \frac{1}{n!} \left(\frac{-i\lambda}{4!} \right)^n$$

If interpretation (1) = interpretation (2)

Any diagram can be thought as product of $\left(\sum_{\text{form}} \text{diagram without bubble} \right) \times (1 + \text{Sum of bubbles})$.

i.e.

Sum of Feynman diagrams = (sum of connected) $\times$ (1 + sum of bubbles)

$$= \sum_{\text{connected}} \cdot \sum (1 + \text{bubbles})$$

The key is to find combinatorial factors which is hard to find if we don't have diagrams. If we had diagram we can follow it to find combinatorial factor.

But exchange of vertices don't generate new diagrams

So, 1) $\frac{1}{(m+n)!} \left(\frac{-i\lambda}{4!} \right)^{m+n} \times (m+n)! \quad \checkmark \text{ factors for exchange of vertices.}$

$$2) \frac{1}{m!} \left(\frac{-i\lambda}{4!} \right)^m \frac{1}{n!} \left(\frac{-i\lambda}{4!} \right)^n \times (m! n!)$$

Counting (1)

of ways of choosing 'm' vertices out of $(m+n)$ is ${}^{m+n}C_m$

$$= \frac{(m+n)!}{m! n!} \quad \left\{ {}^nC_r = \frac{n!}{r! (n-r)!} \right.$$

_____ $\times$ _____

We were after propagator $\Delta(x, x')$ satisfying (we need to find $\Delta(x, x')$)

$$(\partial_x^2 - m^2) \Delta(x, x') = i \delta^4(x - x') \quad (\text{translation invariant})$$

$$\begin{array}{lcl} x & \longrightarrow & x + a \\ x' & \longrightarrow & x' + a \end{array} \quad \text{eq}^n \text{ will look same.}$$

So, this $\Delta(x, x')$ shall only depend upon $|x - x'|$.

Fourier expansion of

$$\Delta(x, x') = \int \frac{d^4 k}{(2\pi)^4} e^{i k \cdot (x - x')} \tilde{\Delta}(k)$$

if we use above eqⁿ

$$\begin{aligned} (\partial_x^2 - m^2) \Delta(x, x') &= \int \frac{d^4 k}{(2\pi)^4} (\partial_x^2 - m^2) e^{i k \cdot (x - x')} \tilde{\Delta}(k) \\ &= \int \frac{d^4 k}{(2\pi)^4} (i^2 k^2 - m^2) e^{i k \cdot (x - x')} \tilde{\Delta}(k) \\ &= \int \frac{d^4 k}{(2\pi)^4} (-k^2 - m^2) e^{i k \cdot (x - x')} \tilde{\Delta}(k) \\ &= i \delta^4(x - x') \quad \text{iff : } \tilde{\Delta}(k) = \frac{i}{-k^2 - m^2} \end{aligned}$$

$$\tilde{\Delta}(k) = \frac{i}{-(k_0^2 + k_i k^i + m^2)} = \frac{i}{-(-k_0^2 + k_i \cdot k_i + m^2)}$$

$$\tilde{\Delta}(k) = \frac{i}{k_0^2 - \vec{k}^2 - m^2} \quad \dots (\eta_{\mu\nu} = (-1, 1, 1, 1))$$

Not defined fully at denominator = 0.

So, it needs to be specified how to deal with singularity.

$$\text{Singularity at } k_0 = \pm \sqrt{\vec{k}^2 + m^2}$$

Ambiguity $\propto e^{i k \cdot (x - x')}$ at $k_0 = \pm \sqrt{\vec{k}^2 + m^2}$ (in x -dependence)

i.e. expression for $\Delta(x, x')$ will change by $e^{i k \cdot (x - x')}$ at $k_0 = \pm \sqrt{\vec{k}^2 + m^2}$

Also $e^{i k \cdot (x - x')}$ satisfies

$$\begin{aligned} (\partial_x^2 - m^2) e^{i k \cdot (x - x')} &= 0 \\ \underline{\text{LHS}} : (-k^2 - m^2) e^{i k \cdot (x - x')} &= (0) \cdot e^{i k \cdot (x - x')} \\ &\downarrow \\ \text{at } k_0 &= \pm \sqrt{\vec{k}^2 + m^2} \end{aligned}$$

So, this ambiguity is just reflecting the fact that this eqⁿ

$$(\partial_x^2 - m^2) \Delta(x, x') = i \delta^4(x - x')$$

does not have unique solⁿ (i.e. $\Delta(x, x')$).

for ex: we can $\Delta \rightarrow \Delta' = \Delta + \text{Const.}$

$$\text{Still } (\partial_x^2 - m^2) \Delta' = i \delta^4(x - x')$$

Greens function is not uniquely defined in minkowaski space.

Shortcut:

$$x^0 - (x')^0 \longrightarrow (x^0 - x'^0) \cdot (1 - i\epsilon)$$

So,

$$\begin{aligned} \Delta(x, x') &= \int \frac{d^4 k}{(2\pi)^4} e^{-i k^0 (x^0 - x'^0) + i \vec{k} \cdot (\vec{x} - \vec{x}')} \tilde{\Delta}(k) \\ &= \int \frac{d^4 k}{(2\pi)^4} e^{-i k^0 (x^0 - x'^0)(1 - i\epsilon) + i \vec{k} \cdot (\vec{x} - \vec{x}')} \tilde{\Delta}(k) \\ &= \int \frac{dk^0}{2\pi} \frac{d^3 k}{(2\pi)^3} e^{-k^0(i+\epsilon)(x^0 - x'^0) + i \vec{k} \cdot (\vec{x} - \vec{x}')} \tilde{\Delta}(k) \\ &\quad \downarrow \text{ will blow up as } k^0 \rightarrow (-\infty). \end{aligned}$$

So to prevent it, we also rotate k^0

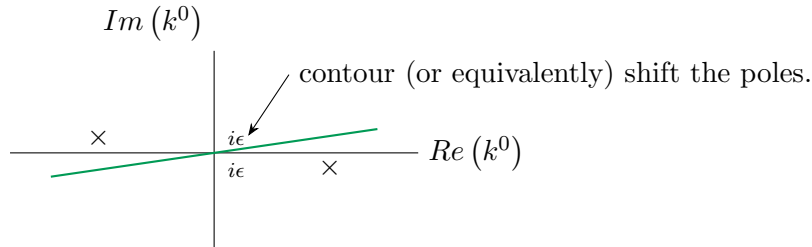
$$k^0 \longrightarrow \frac{k^0}{1 - i\epsilon} = k^0 (1 + i\epsilon) + O(\epsilon^2) \quad \hookrightarrow \text{does it really prevent ?}$$

Such that

$$\begin{aligned} \tilde{\Delta}(k) &= \frac{i}{k'^2 - \vec{k}^2 - m^2} \longrightarrow \frac{i}{k^{02} (1 + i\epsilon)^2 - \vec{k}^2 - m^2} \\ &= \frac{i}{k^{02} (1 + 2i\epsilon) - \vec{k}^2 - m^2} \quad (\text{neglecting } O(\epsilon^2)) \\ &= \frac{i}{(k^0)^2 - \vec{k}^2 - m^2 + 2i\epsilon k^{02}} \\ &= \frac{i}{(k^0)^2 - \vec{k}^2 - m^2 + i\epsilon} \quad \text{redefining } \epsilon. \end{aligned}$$

poles are now shifted

$$k^0 = \begin{cases} \sqrt{\vec{k}^2 + m^2} - i\epsilon \\ -\sqrt{\vec{k}^2 + m^2} + i\epsilon \end{cases}$$



So, Final form of propagator becomes :

$$\tilde{\Delta}_F(k) = \frac{i}{-k^2 - m^2 + i\epsilon} \quad \text{where } \epsilon > 0.$$

In path integral formalism $i\epsilon$ prescription ensures that we get vacuum state to calculate partition function.

$$e^{iHT} |q'\rangle \longrightarrow e^{iHT} \sum_n |n\rangle \langle n|q'\rangle = \sum_n e^{iE_n t} \psi_n(q') |n\rangle$$

if $T \longrightarrow T(1-i\epsilon)$

$$\searrow e^{iHT(1-i\epsilon)} \sum_n |n\rangle \langle n|q'\rangle = \sum_n e^{iE_n T(1-i\epsilon)} \langle n|q'\rangle |n\rangle$$

at $t = -\infty$ only $n = 0$ state survives

$$\sum_n e^{iE_n t} e^{\epsilon E_n T} |n\rangle \langle n|q'\rangle. \quad \checkmark \quad \text{What if } E_0 \neq 0 \text{ does it still survive?}$$

Also $m^2 \rightarrow m^2(1-i\epsilon)$ makes path integral well defined! else we have damped fun. e^{iS} .

Momentum space feynman rules :

Idea is to calculate fourier transform of green's function.

$$\begin{aligned} \tilde{G}(k_1 k_2 \dots k_n) &= \int d^4x_1 d^4x_2 \dots d^4x_n e^{-ik_1 \cdot x_1} e^{-ik_2 \cdot x_2} \dots e^{-ik_n \cdot x_n} \\ &\quad \times G_n(x_1 x_2 \dots x_n) \end{aligned}$$

We have

$$\begin{aligned} \tilde{\phi}(k) &= \int d^4x \phi(x) e^{-ik \cdot x} \\ \phi(x) &= \int \frac{d^4k}{(2\pi)^4} e^{ik \cdot x} \tilde{\phi}(k) \end{aligned}$$

In $Z[J]$, we will replace $\mathcal{D}\phi \rightarrow \mathcal{D}\tilde{\phi}$ there will be some Jacobian (const. but independent of source term $J(x)$ which will cancel out when we divide by $Z[0]$.) $\nearrow$ what will be Jacobian (?)

$$\begin{aligned} Z[J] &= \frac{1}{Z[0]} \int [\mathcal{D}\phi] e^{iS+i \int d^4x \phi(x) J(x)} \\ &= \frac{1}{Z[0]} \int [\mathcal{D}\tilde{\phi}] e^{iS+i \int d^4x J(x) \left(\int \frac{d^4k}{(2\pi)^4} e^{ik \cdot x} \tilde{\phi}(k) \right)} \end{aligned}$$

also

$$S_{\text{free}}[\phi] = \frac{1}{2} \int d^4x (\eta^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - m^2 \phi^2)$$

using $\phi(x) = \int \frac{d^4k}{(2\pi)^4} e^{ik \cdot x} \tilde{\phi}(k)$

$$\begin{aligned} S_{\text{free}}[\tilde{\phi}] &= \frac{1}{2} \int d^4x \int \frac{d^4k}{(2\pi)^4} \frac{d^4k'}{(2\pi)^4} \left[(ik_\mu)(ik'_\nu) \tilde{\phi}(k) \tilde{\phi}(k') \eta^{\mu\nu} e^{ik \cdot x} e^{ik' \cdot x} \right. \\ &\quad \left. - m^2 \tilde{\phi}(k) \tilde{\phi}(k') e^{ik \cdot x} e^{ik' \cdot x} \right] \end{aligned}$$

where $\left\{ d\check{k} = \frac{dk}{2\pi} \right\}$

$$= \frac{1}{2} \int d^4x \int d^4\check{k} d^4\check{k}' \cdot e^{ix \cdot (k+k')} \tilde{\phi}(k) \tilde{\phi}(k') (-k^2 - m^2)$$

solving $\int d^4x$ we get

$$= \frac{1}{2} \int d^4\check{k} d^4\check{k}' (2\pi)^4 \delta^4(k+k') \tilde{\phi}(k) \tilde{\phi}(k') (-k^2 - m^2)$$

solving k' integral we get

$$S_{\text{free}}[\tilde{\phi}] = \frac{1}{2} \int \frac{d^4k}{(2\pi)^4} \underbrace{\tilde{\phi}(k) \tilde{\phi}(-k)}_{\downarrow \text{quadratic term } \textcircled{1}} \left(\textcircled{-k^2 - m^2} \right) \nearrow \text{Inverse of propagator}$$

Similarly

$$\begin{aligned} S_{\text{int}} &= \int d^4x L_{\text{int}} = \int d^4x \frac{\lambda}{4!} \phi^4(x) \\ &= \frac{\lambda}{4!} \int d^4x \int d^4\check{k}_1 d^4\check{k}_2 d^4\check{k}_3 d^4\check{k}_4 e^{i x \cdot (k_1 + k_2 + k_3 + k_4)} \tilde{\phi}(k_1) \tilde{\phi}(k_2) \tilde{\phi}(k_3) \tilde{\phi}(k_4) \end{aligned}$$

solving ' x ' integral

$$S_{\text{int}} = \frac{\lambda}{4!} \int d^4\check{k}_1 d^4\check{k}_2 d^4\check{k}_3 d^4\check{k}_4 (2\pi)^4 \delta^4(k_1 + k_2 + k_3 + k_4) \tilde{\phi}(k_1) \tilde{\phi}(k_2) \tilde{\phi}(k_3) \tilde{\phi}(k_4)$$

the Source Action becomes

$$\begin{aligned} i \int \underbrace{J(x)}_{\downarrow \check{k}'} \underbrace{\phi(x)}_{\downarrow k} d^4x &= i \int d^4x \int d^4\check{k} d^4k' \tilde{J}(k') \tilde{\phi}(k) e^{i x \cdot (k+k')} \\ &= i \int d^4\check{k} d^4\check{k}' \tilde{J}(k') \tilde{\phi}(k) (2\pi)^4 \delta^4(k+k') \\ &= i \int \frac{d^4k}{(2\pi)^4} \tilde{J}(-k) \tilde{\phi}(k) \quad \hookrightarrow \text{linear term } \textcircled{2} \end{aligned}$$

① is quadratic term , ② is linear term , so we complete the square.

It turns out that:

$$\frac{Z_f[J]}{Z_f[0]} = \exp \left(-\frac{1}{2} \int d^4\check{k}_1 d^4\check{k}_2 \Delta_F(k_1, k_2) \tilde{J}(k_1) \tilde{J}(k_2) \right)$$

$$\Delta_F(k_1 k_2) = (2\pi)^4 \delta^4(k_1 + k_2)$$

$$\begin{aligned} G_F(k_1 k_2 \cdots k_n) &= \left(- (2\pi)^4 i \frac{\delta}{\delta \tilde{J}(-k_1)} \right) \left(- (2\pi)^4 i \frac{\delta}{\delta \tilde{J}(-k_2)} \right) \\ &\quad \cdots \left(- (2\pi)^4 i \frac{\delta}{\delta \tilde{J}(-k_N)} \right) \left. \frac{Z[J]}{Z[0]} \right|_{J=0}. \end{aligned}$$